

**DRUG ENFORCEMENT ADMINISTRATION
AVIATION DIVISION
Specifications
TWIN ENGINE TURBOPROP AIRCRAFT**

1.0 OVERVIEW

The U.S. Department of Justice, Drug Enforcement Administration (DEA) has a requirement for new or used Beechcraft King Air 350/360 aircraft. This contract is for the purchase of one or more aircraft. This specification establishes the technical requirements for new Beechcraft King Air B300ER/B300CER and/or used Beechcraft King Air B300/B300ER/B300C/B300CER aircraft to be owned and operated by the DEA as a multi-role aircraft.

1.1 PURPOSE

This contract will provide the Drug Enforcement Administration with new or used Twin Engine Turboprop Aircraft as described in this requirement and will be utilized in the deployment of government personnel and equipment in both foreign and domestic areas.

2.0 GENERAL REQUIREMENTS

The aircraft shall be a new or used commercially available Twin Engine Turboprop Aircraft with a minimum seating for 2 crew members and 8 passengers. The aircraft shall be configured with a cargo door, but standard door will be considered. It is essential that the aircraft have a long endurance operation capability. All factory standard equipment of the manufacturer's commercial model, unless otherwise noted, shall be provided in addition to those items specified herein. These requirements, together with the following specifications, shall constitute the minimum acceptable requirements.

3.0 SPECIFIC REQUIREMENTS

The aircraft, when configured in accordance with this entire specification, shall meet or exceed the specifications in the following paragraphs.

3.1 CERTIFICATION

The aircraft shall be certified in accordance with the U.S. Federal Aviation Administration (FAA), Federal Aviation Regulations Part 23, Airworthiness Standards: Normal, Utility, Acrobatic and Commuter Category Airplanes including day, night, VFR, IFR, and flight into known icing conditions and is compliant with RVSM certification requirements. All options and their installation shall also be FAA approved. The Standard Airworthiness Certificate and any other FAA approvals shall be made available to the government upon delivery.

3.2 AIRCRAFT SPECIFICATIONS

3.2.1 Aircraft Flight Performance Requirements

The following are minimum performance requirements:

1. IFR Range – 4 passengers, no wind, NBAA IFR reserve profile with 100 nm diversion.
 - Maximum Speed Cruise at 35,000 ft. – 1533 nm
 - Maximum Range Cruise at 35,000 ft. – 1715 nm
2. Take-Off Field Length
 - Sea Level, ISA, 15,000 lbs. (MTOW) – 3300 ft.
3. Landing Distance (FAR Part 25)
 - Sea Level, ISA, 15,000 lbs. (MLW) – 2692 ft.
 - At Typical Landing Weight (11,500) – 2415 ft.
4. Rate of Climb (Second Segment, Flaps Up, Gear Up)
 - Two Engines Climb Rate (at MTOW, SL, ISA) – 2700 fpm
 - One Engine Inop. Climb Rate (at MTOW, SL, ISA) – 622 fpm
5. Cruise Speed/Fuel Flow
 - High Speed Cruise, Typical Mid-Cruise Weight, 12,500 lbs.
Cruise Speed – 312 KTAS
Fuel Flow – 773 lbs./hr.
 - Long Range Cruise, Typical Mid-Cruise Weight, 12,500 lbs.
Cruise Speed – 237 KTAS
Fuel Flow – 362 lbs./hr.
6. Cruise Altitude
 - Initial Climb at MTOW – 35,000 ft.
 - Maximum Certified Ceiling – 35,000 ft.
 - Engine-Out Service Ceiling – 21,500 ft.

3.2.2 Aircraft Avionics

The aircraft shall be FAA certified for Instrument Flight Rules (IFR) operations and Reduced Vertical Separation Minimum (RVSM) operation approved with an aircraft specific Minimum Equipment List (MEL).

The following communication/navigation equipment is the government's minimum requirement. A Collins Pro Line Fusion Avionics System or Garmin G1000NXi System to include:

1. Avionics System to include:
 - a. GPS with WAAS/LPV receiver
 - b. VHF Radios
 - c. Navigation Receivers
2. Dual Audio System to include:
 - a. Dual Audio Panels
 - b. Crew Interphone
3. Cockpit Voice Recorder
Solid State Cockpit Voice Recorder
4. Dual Flight Management System to include:
 - a. Moving Map Displays on the Multi-Function Display
 - b. Flight Planning, Direct-To Navigation
 - c. Enroute and Terminal Operations
 - d. Navigational Operations based upon VOR/DME and GPS and WAAS/LVP approach capability and Radius-to-fix legs
5. Electronic Charts to include:
 - a. Geo-referenced electronic navigation charts
 - b. Airport diagrams for surface navigation
6. Surface Management System
The Surface Management System with automated checks and aural advisories to the flight crew and a visual overlay that highlights the target runway on the airport chart display.
7. Distance Measuring Equipment (DME)
Scanning DME/DME input capability for the Flight Management System.
8. Engine Indicating and Crew Alerting System
Electronic display of primary engine operating information with alerts.
9. Flight Guidance System to include:
 - a. Flight Director
 - b. Autopilot
 - c. Yaw Damper
 - d. Automatic Pitch Trim
10. Air Data System
An Air Data System that supplies digital output signals (airspeed and altitude) to the AHRS, transponders and Flight Guidance System and shall be RVSM capable.

11. Attitude/Heading Reference System (AHRS)
An AHRS that provides attitude, heading and flight dynamics information to the flight control and display system.
12. Transponders with ADS-B Out
Dual Collins TDR-94D Mode A/C/S Transponders with ADS-B Out and shall be compliant with the DO-260B standard.
13. Weather Radar
Collins Multi-Scan RTA-4112 Weather Radar System.
14. Radar Altimeter
One each Collins ALT-4000 Radar Altimeter.
15. Traffic Collins Avoidance System (TCAS II)
Collins TTR-4100 System.
16. Integrated Terrain Awareness Warning System (iTAWS)
Collins iTAWS System that shall include Class A Terrain Awareness and Warning System displayed on the PFD and MFD.
17. Synthetic Vision Technology
Collins Pro Line Fusion Synthetic Vision Technology.
18. Emergency Location Transmitter
Artex C406N Emergency Locator Transmitter.
19. Standby Instrumentation
L3 Communications GH-3900 Electronic Standby Instrument System.
20. Onboard Maintenance System
An Onboard Maintenance System that provides data to maintenance technicians to be used for aircraft maintenance.

3.3 Aircraft Configuration

The aircraft shall be a new, twin-engine turboprop with a large, pressurized cabin/cargo area of no less than 350 cubic feet. The aircraft shall have crew seating of no less than 2 and passenger seating of no less than 8. The passenger seats shall be easily removable to allow the aircraft to be quickly configured for cargo operations. The aircraft shall have an interior baggage/storage space of no less than 53 cubic feet and a weight capacity of no less than 550 lbs. The aircraft with a cargo door shall have a door opening of no less than 52 inches by 49 inches.

3.4 Additional Aircraft Equipment

The aircraft shall have the following additional equipment installed. All additional equipment and their installations shall be FAA approved.

1. Beechcraft King Air 350ER Extended Range Fuel Tanks and Heavy Weight Certification Option
2. 24 Inch Extended Pedestal
3. Aft Cabin Stub Partition/Privacy Curtain/Baggage Web

3.5 Aircraft Condition/Age

The aircraft shall be a new or used, commercially available twin engine turboprop aircraft with no more than 15 hours total time on the airframe and no more than 15 hours total time on either engine for new aircraft. Used aircraft shall be manufactured in 2016, or newer, less than 5,000 hours aircraft and engine total time, less than 4,000 total landings/cycles, and must have at least 50% operational time/landings/cycles remaining before any airframe/engine/component overhaul or life-limited part replacement.

3.6 Technical Publications

The contractor shall provide technical publications required for operation and maintenance of the aircraft and installed systems to include aircraft maintenance manuals, aircraft specific wiring manual, engine maintenance manuals and illustrated parts manuals for the aircraft and engines. All aircraft technical manuals shall be delivered with a one-year revision subscription included.

4.0 Training

The contractor shall provide the following training for each aircraft purchased:

1. Pilot training, type rating, initial course, 2 each.
2. Maintenance training initial course, 1 each.

5.0 Delivery

Delivery of a new aircraft shall be within twelve (12) months of aircraft order. Delivery of used aircraft shall be within six (6) months of aircraft purchase. The government will take delivery of the aircraft at the contractor's facility, or a mutually agreed upon location.

6.0 Government Furnished Equipment

None.

7.0 Point of Contact

The contractor shall designate a single person as a point of contact for all communications with the Government regarding this contract.

8.0 Warranty

New aircraft:

1. Five years on Aircraft structure (fuselage, empennage, wing and control surfaces).
2. Two years or 1,200 operating hours on airframe parts, components and equipment.
3. Five years for Avionics.
4. Five years or 2500 hours on engines.
5. 36 months or 2000 hours on propellers.

Used aircraft:

1. The contractor shall warrant the aircraft meets all requirements of this contract.
2. All remaining factory or manufacturer's warranty shall be transferred to the Government

9.0 Applicable Standards and Regulations

14 CFR- U. S. Federal Aviation Administration, Federal Aviation Regulations